

Francesco Di Lucia

Software Engineer | AI Applications & Integrations

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Profile

Computer Engineering MSc graduate with hands-on projects in tool-using LLM agents, local RAG applications and backend integrations. Combines Python and machine-learning experience with API development, systematic debugging and technical documentation. Looking to build useful AI applications while growing within an experienced engineering team.

Selected Projects

Bounded Agent Lab | Agent Runtime Prototype GitHub

- Built a Python tool-using agent prototype with OpenAI Agents SDK, OpenRouter and MCP, with separate modes for offline code execution and public web research.
- Implemented Docker isolation, human-approved file exports, token and estimated-cost budgets, usage logging and regression tests for execution limits and failure handling.

LLittlebrain | Local LLM Chat & Document RAG GitHub

- Built a full-stack application with React/TypeScript, FastAPI and Ollama, with streaming responses and persistent conversation history in SQLite.
- Implemented document ingestion, overlapping text chunks, local embeddings and ChromaDB retrieval to provide relevant document context and source references for answers.

Car Rental Microservices Platform GitHub

- Contributed REST API integrations to a Kotlin/Spring Boot system, including Keycloak/OAuth2 authentication and Kafka event flows between services.

Experience

Research Intern / Master's Thesis Researcher Jul 2025 – Jul 2026

DAUIN, Politecnico di Torino | DRONUTS project GitHub

- Developed an end-to-end Python/PyTorch pipeline to classify healthy and diseased plants from UAV multispectral imagery, building a dataset of 3,643 image patches.
- Compared band alignment, normalization, vegetation segmentation, custom CNNs and transfer learning; used group-aware data splits and documented evaluation results and limitations.
- Investigated whether cleaner tree crops improved classification; used the unexpected results to reassess label ambiguity and explain technical trade-offs to supervisors.

Computer Science Teaching Assistant | Istituto Agnelli Sep 2021 – Jul 2023

- Supported students in Python, C++, Java, JavaScript, algorithms and OOP, adapting explanations and debugging guidance to different experience levels.

Education

MSc in Computer Engineering, AI and Data Analytics Oct 2023 – Jul 2026

Politecnico di Torino | Thesis: *Automatic Classification of Healthy / Diseased Plants using Multispectral Images*

BSc in Computer Engineering | Politecnico di Torino Oct 2017 – Jul 2023

Technical Skills

AI & LLM Applications: PyTorch, scikit-learn, OpenCV, RAG, Ollama, ChromaDB, MCP, OpenAI Agents SDK, OpenRouter

Software & Data: Python, TypeScript/JavaScript, React, FastAPI, Kotlin, Java, C, C++, Spring Boot, REST APIs, SQL, PostgreSQL, SQLite, Kafka, OAuth2/Keycloak, NumPy, pandas

Tools: Git/GitHub, Docker, Linux/Unix, unittest, technical documentation

Languages: Italian (native), English (C2)